

Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups

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1 Abstract

We study two metadata channels that persist even when payload privacy is perfect: **(i)** observable *termination / stop time*, and **(ii)** observable *destination / contact traces* over repeated operations. We develop a proof-oriented **compiler calculus**: small, auditable transformations that export quantitative contracts (TV, pointwise, KL/MI) and can be composed into end-to-end leakage guarantees.

For stop times, we analyze a simple ε -tunable **mixture compiler** and show it is optimal *within the mixture class*. We then show that broader **kernel-feasible** padding rules can be strictly better: we formulate globally optimal kernel-feasible padding as a linear program (**Kernel-TV-Pad**), give explicit separating instances and parametric families where mixtures are arbitrarily suboptimal, and provide machine-checkable primal-dual certificates.

For contact traces, we combine **public scheduling** (PSC), which makes the set of coarse destinations a deterministic function of public time, with **mixture equalization** (MC-EQ), which equalizes within each scheduled set via one-real-among-w mixing. Our main composition theorems convert these module contracts into (i) transcript-by-transcript **posterior** guarantees near $1/K$ and (ii) **MI/Fano** identification bounds, together with a reproducible parameter-selection worksheet and figures.

2 1. Motivation, leakage channels, and notation

Overlay lookup protocols (including many DHT-like designs) can hide *what* is being queried while still leaking *how the query unfolds*. This paper treats two particularly stubborn metadata channels: - **Stop time**: when the observable protocol terminates. - **Contact trace / marks**: which coarse destinations (e.g., shards/committees) are touched at each time slot. Our goal is a **compiler calculus**: small, auditable transformations that (i) fit implementability constraints (causality, feasibility), and (ii) admit quantitative leakage bounds that compose. We emphasize **two complementary leakage regimes**: - **Posterior-first** (near- $1/K$ posteriors): protect against rare transcripts that spike the posterior. - **MI/Fano** (average information): useful when only

high-confidence identification needs to be ruled out. A transcript consists of: - a stopping time T (termination-only leakage), or - a contact-mark sequence $O_{1:T}$ (e.g., “which shard/committee was contacted in slot t ”), together with a public schedule. We write $\text{TV}(P, Q)$ for total-variation distance and $D_{\text{KL}}(P \| Q)$ for KL divergence (nats). We use $I(C; O)$ for MI between a hidden client/profile variable C and observations O . —

2.1 1.1 A compiler-calculus interface (modules and contracts)

To keep the end-to-end arguments modular, we treat each transformation as exporting a small **contract** that can be consumed by a composition theorem. Informally: - the **trace compiler** exports per-slot deviation bounds relative to a secret-independent reference kernel, - the **stop-time compiler** exports a TV / posterior-friendly bound for the padded termination distribution, - the **composition theorem** consumes these bounds to control either worst-case **posteriors** or average **MI/Fano** risk. We summarize the main modules and their “typed” outputs in Table 1. **Table 1 (Module contracts used by the composition theorems).**

Module	What it does	Exported contract (parameters)	Consumed by
PSC (public scheduling)	Makes the action-set schedule J_t a deterministic function of public time; collapses shard/committee marks to within-set choice.	Public sets J_t ; (optionally) a support-containment guarantee for KL/MI accounting (Remark 3.1).	Sections 3–6; Theorems 10.1–10.2.
MC-EQ (mixture equalization)	Within a scheduled set J_t of size m , mixes one secret-dependent action with $w - 1$ covers from a secret-independent kernel $K_t(\cdot h)$.	Per-slot deviations relative to K_t : $\text{TV} \leq 1/w$ and pointwise $ P(y c, h) - K_t(y h) \leq 1/w$ (Lemma 4.2).	Theorem 7.2; Theorems 10.1–10.2.
StopTimePad (Kernel-TV-Pad)	Kernel-feasible padding $\Phi(t s)$ supported on $t \geq s$ that minimizes expected padded stop time under a pairwise TV budget.	Pairwise stop-time privacy: $\text{TV}(P_c, P_{c'}) \leq \tau$ for all c, c' .	Section 2; Theorem 10.2 (MI/Fano path).
StopTimePad (PosteriorStopTime-Pad)	Stop-time padding tuned for posterior-first composition: enforces an ℓ_∞ -star constraint around a reference R with a probability floor.	Existence of R with floor κ_T and additive deviations δ_T : $ P_c(t) - R(t) \leq \delta_T$, $R(t) \geq \kappa_T$.	Proposition 2.15; Theorem 10.1 (posterior-first path).

Module	What it does	Exported contract (parameters)	Consumed by
Composition	Converts per-slot contracts (trace + stop time) into transcript-level leakage bounds.	(i) Posterior guarantee (Theorem 10.1), (ii) MI/KL bound for Fano (Theorem 10.2).	End-to-end results + design worksheet (Section 10.4).

This interface perspective is what lets Option B (“compiler calculus”) scale: adding a new compiler is useful whenever it exports a contract that composes.

2.2 1.2 Threat model and scope

We assume an adversary who observes the **public schedule** (PSC), the per-slot **contact marks** (e.g., shard/committee identifiers contacted in each slot), and the final **stop time**. Payload contents are protected by standard cryptography (e.g., PIR inside a shard, encryption, authentication), and real versus dummy messages are assumed computationally indistinguishable at the payload layer. Our focus is on *metadata* that remains: the observable sequence of marks and the termination time. **What we cover.** - Coarse destination leakage (marks) at the granularity of scheduled action sets. - Stop-time leakage under implementable (causal, monotone) padding rules. - Two leakage regimes: worst-case **posterior spikes** (near- $1/K$ posteriors) and average **MI/Fano** leakage. **What we do not cover.** We do not model packet-level timing within a slot, size/shape leakage, routing-layer side channels, churn-induced linkability, or active attacks that break the payload-layer indistinguishability assumption. These can be layered on top, but are orthogonal to the compiler calculus developed here.

2.3 1.3 Main results at a glance

The paper is organized around a modular pipeline; the most important results can be read as *typing rules* for modules and as *composition theorems* that consume their contracts. **Stop-time compilers (Section 2).** - Mixture compilers achieve $\text{StopTime-IND}_\epsilon$ with a linear privacy-overhead dial and are tight *within the mixture class*. - Kernel-feasible padding is strictly more powerful: we give explicit separations, an LP formulation (**Kernel-TV-Pad**) with a dual/KKT form (certificate-friendly), and a closed-form parametric family where mixtures are **arbitrarily suboptimal** (gap $\Omega(n)$); see Proposition 2.2 and Theorem 2.8. - For K -ary stop-time hiding, we formalize **pairwise** vs **TV-star** constraints, prove a tight TV geometry lemma (Lemma 2.13), and provide machine-checkable primal-dual certificates for strict gaps (Section 2.9). **Trace compilers and composition (Sections 3–10).** - PSC collapses shard-level marks to within-set choices. - MC-EQ provides a tunable equalization knob w and exports both TV and pointwise deviation contracts (Lemma 4.2). - The flagship **compiler calculus theorems** (Theorems 10.1 and 10.2) compose PSC + MC-EQ + stop-time padding into (i) transcript-by-transcript posterior guarantees and (ii) MI/Fano guarantees, with a worked parameter-selection worksheet (Section 10.4). **Figure 1 (pipeline).** A one-page diagram in Appendix A summarizes the module contracts and how they feed into Theorems 10.1–10.2.

2.4 1.4 Notation quick reference

We use the following symbols repeatedly; the goal is to keep the later proofs “type-checkable” at a glance. - C : hidden client/profile/secret, typically uniform on $[K]$. - K : number of possible

secrets/profiles. - T : (padded) observable stop time; S : (un-padded) success time. - n : discrete time horizon for stop-time padding examples. - O_t : observable mark in slot t (e.g., which action within the scheduled set was contacted). - J_t : scheduled action set in slot t (public under PSC), with $|J_t| = m$. - m : scheduled-set size (number of actions available in a slot). - w : equalization fanout / cover multiplicity (“one real among w ”), with $\rho := 1/w$. - τ : TV privacy budget for stop-time distributions (pairwise $\text{TV}(P_c, P_{c'}) \leq \tau$). - δ : generic per-step deviation parameter (TV or pointwise), with floors κ when converting to KL. - $\text{TV}(P, Q)$: total variation; $D_{\text{KL}}(P\|Q)$: KL divergence (nats); $I(C; O)$: mutual information.

2.5 1.5 Reproducibility

This draft is accompanied by small deterministic scripts that reproduce every numerical claim and primal–dual certificate. To run the full suite, execute:

```
bash reproduce_all_v67.sh
```

Run the command from the directory that contains the scripts (e.g., the artifact root). The runner is path-robust and will regenerate Figures 2–3.

Typical runtime on a laptop is under a minute; most certificates complete in seconds.

Appendix C lists the mapping from claims to scripts.

2.6 1.6 How to read this paper

- For **stop-time padding** (timing-only leakage), start with Section 2. Sections 2.4–2.10 contain the LP formulations, separations from mixture compilers, and primal–dual certificates.
- For **contact traces** (who is contacted when), read Sections 3–6: PSC collapses shard-mark leakage, MC-EQ provides the per-slot equalization knob, and we relate per-step mixing to the hypotheses of the sequential theorems.
- For an **end-to-end privacy guarantee**, jump to Theorem 10.1 (posterior-first, near- $1/K$ posteriors) and Theorem 10.2 (MI/Fano, high-confidence identification risk). Section 10.4 gives a worked design worksheet.

3 2. Stop-time leakage: StopTime-IND and mixture compilation

3.1 2.1 StopTime-IND $_{\epsilon}$

In the stop-time-only leakage model, the adversary observes only the stop time T . StopTime-IND $_{\epsilon}$ is equivalent to $\|P_0 - P_1\|_{\text{TV}} \leq 2\epsilon$.

3.2 2.2 Mixture compiler and tightness within the mixture class

A mixture compiler $C_{\lambda, Q}$ (probability λ stop at true success time S ; otherwise stop at secret-independent $T_Q \sim Q$ with $T_Q \geq S$ a.s.) yields $\|P_0 - P_1\|_{\text{TV}} \leq \lambda$, hence choosing $\lambda = 2\epsilon$ gives StopTime-IND $_{\epsilon}$. **Lemma 2.1 (tightness within the timeout-mixture class).** Consider the following timeout-mixture padding rule: fix a secret-independent distribution Q supported on times t that are always feasible (e.g., a deterministic public timeout at the horizon n), and output $T = S$ with probability λ and $T \sim Q$ with probability $1 - \lambda$. Then for any two worlds b, b' ,

$$\text{TV}(P_b, P_{b'}) = \lambda \text{TV}(S_b, S_{b'}),$$

so in particular $\text{TV}(P_b, P_{b'}) \leq \lambda$ and this bound is **tight** whenever $\text{TV}(S_b, S_{b'}) = 1$. Consequently, to satisfy $\text{StopTime-IND}_\varepsilon$ against worst-case pairs in this class one must have $\lambda \leq 2\varepsilon$, and choosing $\lambda = 2\varepsilon$ is optimal.

Proof. By construction $P_b = (1 - \lambda)Q + \lambda S_b$, so $P_b - P_{b'} = \lambda(S_b - S_{b'})$ and TV scales linearly. $\square$

3.3 2.3 Global optimal stop-time padding remains open

4 Beyond the timeout-mixture class, the design space includes general kernel-feasible paddings (Section 2.4) and, more broadly, online/stateful padding rules. Characterizing globally optimal padding under additional constraints (unknown or drifting success-time laws, continuous time, tail constraints, or adversarial environments) remains an open direction.

4.1 2.4 Kernel-feasible padding can beat mixture compilers (new)

The mixture compiler of Section 2.2 is optimal *within its mixture class*, but it does not exhaust all implementable stop-time padding rules. In discrete time, an implementable padding rule can be modeled as a Markov kernel $\Phi(t \mid s)$ supported on $t \geq s$, where s is the (local) success time.

Definition 2.1 (kernel-feasible padding). Fix a finite horizon $\{1, \dots, n\}$. A padding kernel is a stochastic matrix Φ with entries $\Phi(t \mid s) \geq 0$ satisfying

$$\sum_{t=s}^n \Phi(t \mid s) = 1 \quad \text{and} \quad \Phi(t \mid s) = 0 \text{ for } t < s.$$

Given a success time $S = s$, the padded stop time T is sampled as $T \sim \Phi(\cdot \mid s)$. For world $b \in \{0, 1\}$ with success-time law S_b , the induced stop-time law is

$$P_b(t) = \sum_{s=1}^n S_b(s) \Phi(t \mid s).$$

We call such a padding rule **kernel-feasible**. Mixture compilers are a (strict) subclass that enforce an output form $P_b = (1 - \lambda)Q + \lambda S_b$ for a single secret-independent distribution Q (chosen to permit a coupling with $T \geq S$ almost surely) and a parameter $\lambda \in [0, 1]$. **Proposition 2.2 (explicit separation).** There exist success-time laws S_0, S_1 and a $\text{StopTime-IND}_\varepsilon$ level ε for which a kernel-feasible padding achieves $\text{StopTime-IND}_\varepsilon$ with strictly smaller expected stop time than any mixture compiler. *Example and proof.* Take $n = 6$ and

$$S_0 = (0.706, 0.033, 0, 0.028, 0.220, 0.013), \quad S_1 = (0.006, 0.474, 0.026, 0.120, 0.342, 0.032)$$

over $\{1, \dots, 6\}$. One checks that $\text{TV}(S_0, S_1) = 0.7$, $\mathbb{E}[S_0] = 2.062$, $\mathbb{E}[S_1] = 3.414$, and that S_1 stochastically dominates S_0 (tail-by-tail). Fix $\varepsilon = 0.1$, i.e. target $\text{TV}(P_0, P_1) \leq 2\varepsilon = 0.2$. (a)

Kernel-feasible padding. Let Φ be the identity for $s \in \{2, \dots, 6\}$, and for $s = 1$ set

$$\Phi(1 \mid 1) = \frac{2}{7}, \quad \Phi(2 \mid 1) = \frac{441}{700}, \quad \Phi(3 \mid 1) = \frac{26}{700}, \quad \Phi(4 \mid 1) = \frac{33}{700}.$$

Then $T \geq S$ always. The induced laws are

$$P_0 \approx (0.201714, 0.477780, 0.026223, 0.061283, 0.220000, 0.013000),$$

$$P_1 \approx (0.001714, 0.477780, 0.026223, 0.120283, 0.342000, 0.032000),$$

which satisfy $\text{TV}(P_0, P_1) = 0.2$ (hence $\text{StopTime-IND}_{0.1}$). The average expected stop time is

$$\frac{1}{2}(\mathbb{E}[T \mid 0] + \mathbb{E}[T \mid 1]) \approx 3.039074.$$

(b) **Mixture lower bound.** For any mixture compiler, $P_b = (1 - \lambda)Q + \lambda S_b$, so

$$\text{TV}(P_0, P_1) = \lambda \text{TV}(S_0, S_1) = 0.7\lambda.$$

Thus $\lambda \leq 2/7$ is necessary to meet $\text{TV}(P_0, P_1) \leq 0.2$. Moreover, since S_1 stochastically dominates S_0 , any Q that can be coupled to satisfy $T \geq S$ in both worlds must stochastically dominate S_1 , giving $\mathbb{E}[Q] \geq \mathbb{E}[S_1] = 3.414$. Therefore the average expected stop time obeys

$$\frac{1}{2}(\mathbb{E}[T \mid 0] + \mathbb{E}[T \mid 1]) = (1 - \lambda)\mathbb{E}[Q] + \lambda \frac{\mathbb{E}[S_0] + \mathbb{E}[S_1]}{2} \geq (1 - \lambda)\mathbb{E}[S_1] + \lambda \frac{\mathbb{E}[S_0] + \mathbb{E}[S_1]}{2}.$$

The right-hand side is minimized by the largest feasible $\lambda = 2/7$ and equals ≈ 3.220857 . Hence every mixture compiler incurs average expected stop time at least 3.220857, strictly larger than the kernel-feasible padding above. This separation shows that letting the padding kernel depend on the realized success time can strictly reduce overhead at a fixed StopTime-IND level. In particular, “mixture optimality” does not extend beyond the mixture class.

4.2 2.5 Optimal kernel-feasible padding is a linear program (new)

Section 2.4 exhibited a hand-designed kernel that beats every mixture compiler at the same StopTime-IND level. This is not an accident: in discrete time, optimizing kernel-feasible padding under a TV constraint is a **linear program**. **Problem 2.3 (Kernel-TV-Pad).** Fix a horizon $\{1, \dots, n\}$, success-time laws S_0, S_1 on $\{1, \dots, n\}$, and a TV budget $\tau \in [0, 1]$. Choose a kernel-feasible padding matrix $\Phi(t \mid s)$ (Definition 2.1) to minimize the *average* expected padded stop time

$$\min_{\Phi} \frac{1}{2}(\mathbb{E}[T \mid 0] + \mathbb{E}[T \mid 1]) \quad \text{s.t.} \quad \text{TV}(P_0, P_1) \leq \tau,$$

where $P_b = S_b \Phi$ are the induced stop-time laws. The constraint $\text{TV}(P_0, P_1) \leq \tau$ can be linearized by introducing auxiliary variables $d_t \geq 0$ such that

$$d_t \geq P_0(t) - P_1(t), \quad d_t \geq -(P_0(t) - P_1(t)), \quad \sum_{t=1}^n d_t \leq 2\tau.$$

Since $P_b(t) = \sum_{s \leq t} S_b(s) \Phi(t \mid s)$ is linear in Φ , all constraints and the objective are linear. **Proposition 2.4 (LP characterization and sparsity).** Kernel-TV-Pad is a linear program in the variables $\{\Phi(t \mid s) : 1 \leq s \leq t \leq n\}$ and $\{d_t\}_{t=1}^n$. Therefore an optimal solution exists at an extreme point (basic feasible solution). In particular, there is an optimal solution with at most $O(n)$ nonzero variables; one crude bound from the standard “#basic variables $\leq$ #constraints” principle is that some optimum uses at most $3n + 1$ positive variables across (Φ, d) . *Proof sketch.* After linearization, the feasible region is a bounded polytope defined by n row-sum equalities, $2n$ linear inequalities enforcing $|P_0(t) - P_1(t)| \leq d_t$, one TV-budget inequality, and nonnegativity. Linear objectives over polytopes attain minima at extreme points, and basic feasible solutions have support bounded by the number of active constraints. **Proposition 2.5 (Kernel-TV-Pad dual and KKT interpretation).** Let $\delta_s := S_0(s) - S_1(s)$ and $w_s := \frac{1}{2}(S_0(s) + S_1(s))$. Consider the linearized Kernel-TV-Pad LP from Problem 2.3. A standard dual for the $\leq$ -formulation (Problem 2.3 with

$A_{ub}x \leq b_{ub}$ and $x \geq 0$) uses variables: - $y_s \in \mathbb{R}$ for the row-sum equalities $\sum_{t \geq s} \Phi(t | s) = 1$, - $\alpha_t, \beta_t \geq 0$ for the two inequalities $(P_0(t) - P_1(t)) \leq d_t$ and $-(P_0(t) - P_1(t)) \leq d_t$, - $\gamma \geq 0$ for the TV-budget constraint $\sum_t d_t \leq 2\tau$, and can be written as:

$$\max \sum_{s=1}^n y_s - 2\tau \gamma \quad \text{s.t.} \quad y_s - \delta_s (\alpha_t - \beta_t) \leq w_s t \quad \forall 1 \leq s \leq t \leq n, \quad \alpha_t + \beta_t \leq \gamma \quad \forall t.$$

Moreover, for any primal-dual optimal pair, complementary slackness implies: - if $\Phi(t | s) > 0$ then $y_s - \delta_s (\alpha_t - \beta_t) = w_s t$; - if $d_t > 0$ then $\alpha_t + \beta_t = \gamma$; - if $\gamma > 0$ then the TV constraint is tight: $\sum_t d_t = 2\tau$. *Interpretation.* γ is the shadow price of tightening the TV budget, and $q_t := \alpha_t - \beta_t$ acts as a signed “price schedule” over output times t . Complementary slackness explains why optimal kernels are typically sparse and randomize only on times t where the reduced cost is zero.

Certificate note. Since Kernel-TV-Pad is an LP, a dual feasible solution matching the primal objective value certifies optimality. For the $n = 6$ separation in Proposition 2.2, our solver returns a primal-dual pair with primal-dual gap $< 10^{-15}$, giving a numerical certificate of optimality for that instance. **Computational note.** Kernel-TV-Pad can be solved efficiently for moderate n by off-the-shelf LP solvers; in our experiments, optimal kernels are typically sparse and randomize on only a few success-time values. We provide a reproducibility script for the LP and a separate script that extracts and checks dual certificates (including complementary slackness).

4.2.1 2.5.1 A primal-dual certificate for the $n = 6$ separation (new)

For the explicit $n = 6$ instance in Proposition 2.2 with $\tau = 0.2$, the LP optimum is attained by the kernel

$$\Phi(\cdot | 1) = (2/7, 441/700, 26/700, 33/700, 0, 0), \quad \Phi(t | s) = \mathbf{1}\{t = s\} \text{ for } s \geq 2,$$

which induces $P_0 = (0.201714286, 0.477780000, 0.026222857, 0.061282857, 0.220000000, 0.013000000)$ and $P_1 = (0.001714286, 0.477780000, 0.026222857, 0.120282857, 0.342000000, 0.032000000)$, with $\text{TV}(P_0, P_1) = 0.2$ and objective $\frac{1}{2}(\mathbb{E}[T | 0] + \mathbb{E}[T | 1]) \approx 3.039074286$. A dual certificate with $\gamma = 267/350$,

$$\alpha = (267/350, 89/175, 0, 0, 0, 0), \quad \beta = (0, 89/350, 89/350, 267/350, 267/350, 267/350),$$

(and a feasible $y \in \mathbb{R}^6$ reported in the script output) achieves the same dual objective value to machine precision. The script `stop_time_dual_cert_v16.py` checks dual feasibility and complementary slackness (max violation $\approx 10^{-16}$) and prints both the primal kernel and dual variables.

4.3 2.6 Separation families and empirical scaling (new)

Section 2.4 gives a fully explicit $n = 6$ separation. To test whether this is a corner case, we solved Kernel-TV-Pad and the best feasible mixture baseline across random instances (Dirichlet draws) using an LP solver. **Proposition 2.6 (a larger explicit instance, found by LP search).** For $n = 10$ and $\tau = 0.2$, the following pair of success-time laws achieves a gap of about 0.4635 in *average expected stop time* between the best mixture compiler and the optimal kernel-feasible padding:

$$S_0 \approx (0.147285, 0.057829, 0.363867, 0.036741, 0.155830, 0.044686, 0.008706, 0.030916, 0.063848, 0.090290), \\ S_1 \approx (0.003472, 0.002795, 0.000898, 0.414900, 0.023744, 0.093175, 0.035323, 0.086139, 0.078840, 0.260714).$$

In this instance the LP optimum has average ≈ 5.8119 , the mixture optimum has average ≈ 6.2754 , and $\text{TV}(P_0, P_1) = 0.2$. **Proposition 2.7 (gaps can be > 0.6 at moderate horizons).** For $n = 15$

and $\tau = 0.2$, an LP search over 400 random trials found an instance with gap ≈ 0.6861 between the mixture optimum and the kernel-feasible optimum, while still satisfying $\text{TV}(P_0, P_1) = 0.2$. *Status.* These larger instances are obtained by an automated search and are **machine-certified optimal** (primal–dual KKT checks) by the scripts listed in Appendix C. They illustrate that mixture suboptimality persists beyond the smallest toy example and can grow with horizon; Section 2.7 then gives an explicit parametric family with closed-form optima and a linear-in-horizon gap. We provide such a family in Section 2.7.

4.3.1 2.6.1 Optimal kernels can randomize on multiple success-time rows

The closed-form family in Section 2.7 exhibits optimal kernels that randomize only on the earliest success time. However, Kernel-TV-Pad can produce optima that randomize on **multiple** success-time values. This matters for implementers: an optimal padding rule may need to treat several success-time regimes differently. **Proposition 2.7.1 (a two-row randomization example).** Let $n = 6$, $\tau = 0.2$, and

$$S_0 = (0.2795072756, 0.0596474087, 0.5803830930, 0.0743372489, 0.0059209143, 0.0002040595),$$

$$S_1 = (0.4861652867, 0.3024844985, 0.0949222891, 0.1079791870, 0.0066330131, 0.0018157256).$$

The Kernel-TV-Pad optimum induces stop-time laws P_0, P_1 satisfying $\text{TV}(P_0, P_1) = \tau$ and is attained by a kernel that randomizes on **two** success-time rows ($s = 1$ and $s = 3$), e.g.

$$\Phi(\cdot \mid 1) = (0.9677824683, 0, 0.0322175317, 0, 0, 0),$$

$$\Phi(\cdot \mid 3) = (0, 0, 0.9259143009, 0.0692989791, 0.0014668514, 0.0033198687),$$

with all other rows deterministic ($\Phi(t \mid s) = \mathbf{1}\{t = s\}$ for $s \notin \{1, 3\}$). The resulting average expected padded stop time is $\frac{1}{2}(\mathbb{E}[T \mid 0] + \mathbb{E}[T \mid 1]) \approx 2.39349$. *Verification.* The script `stop_time_multirow_cert_v47.py` solves Kernel-TV-Pad for this instance and verifies KKT conditions using HiGHS dual multipliers (primal–dual gap and complementary slackness residuals $\lesssim 10^{-15}$).

4.4 2.7 A closed-form parametric separation family (new)

We next give an explicit family S_0, S_1 for which (i) the kernel-feasible optimum is attained by a simple closed-form padding kernel, (ii) the optimal mixture compiler can be computed in closed form, and (iii) the **mixture-vs-kernel gap grows linearly in the horizon**. Fix a horizon $n \geq 3$, a TV budget $\tau \in (0, 1)$, and a parameter $p \in (\tau, 1]$. Consider the success-time laws on $[n] = \{1, \dots, n\}$

$$S_0 := p \delta_1 + (1 - p) \delta_n, \quad S_1 := (p - \tau) \delta_2 + (1 - p + \tau) \delta_n.$$

Then $\text{TV}(S_0, S_1) = p$. Moreover S_1 first-order stochastically dominates S_0 (it shifts mass later).

Theorem 2.8 (closed-form separation; linear gap). For the pair S_0, S_1 above and TV budget τ : 1. **(Kernel-feasible optimum)** The optimal solution of Kernel-TV-Pad keeps all rows $s \geq 2$ as identity and randomizes only on row $s = 1$:

$$\Phi(1 \mid 1) = \tau/p, \quad \Phi(2 \mid 1) = 1 - \tau/p,$$

with $\Phi(t \mid s) = \mathbf{1}\{t = s\}$ for all other (s, t) . The induced stop-time laws are

$$P_0 = \tau \delta_1 + (p - \tau) \delta_2 + (1 - p) \delta_n, \quad P_1 = (p - \tau) \delta_2 + (1 - p + \tau) \delta_n,$$

which satisfy $\text{TV}(P_0, P_1) = \tau$. The resulting **average expected stop time** equals

$$\text{OPT}_{\text{kern}} = \mathbb{E}[S_1] - \frac{(n-1)\tau}{2} = 2(p-\tau) + \frac{\tau}{2} + n\left(1-p + \frac{\tau}{2}\right).$$

2. (**Mixture-compiler optimum**) Any mixture compiler $C_{\lambda, Q}$ yields $\text{TV}(P_0, P_1) = \lambda \text{TV}(S_0, S_1) = \lambda p$, hence feasibility forces $\lambda \leq \tau/p$. Since S_1 stochastically dominates S_0 , any feasible Q must stochastically dominate S_1 , so $\mathbb{E}[Q] \geq \mathbb{E}[S_1]$ with equality at $Q = S_1$. The mixture optimum takes $\lambda = \tau/p$ and $Q = S_1$, giving

$$\text{OPT}_{\text{mix}} = \mathbb{E}[S_1] - \frac{\tau}{2} - \frac{(n-2)\tau^2}{2p}.$$

3. (**Gap**) The suboptimality of the mixture class is explicit:

$$\text{OPT}_{\text{mix}} - \text{OPT}_{\text{kern}} = \frac{(n-2)\tau}{2} \left(1 - \frac{\tau}{p}\right) > 0.$$

Proof sketch. (1) Since $P_1(1) = 0$, TV-feasibility implies $P_0(1) \leq \tau$. But $P_0(1) = S_0(1)\Phi(1 | 1) = p\Phi(1 | 1)$, so any feasible kernel must satisfy $\Phi(1 | 1) \leq \tau/p$. Moving mass in row 1 to earlier output times decreases the objective, so optimality forces $\Phi(1 | 1) = \tau/p$. The remaining mass on row 1 is best placed at time 2 (the earliest time that also matches $P_1(2) = p - \tau$), yielding the claimed $\{1, 2\}$ -supported row. (2) and (3) follow from the mixture TV scaling and the fact that stochastic dominance forces $Q \succeq S_1$. $\square$ **Example.** Take $n = 6$, $\tau = 0.2$, and $p = 0.9$. Then $\text{OPT}_{\text{mix}} - \text{OPT}_{\text{kern}} \approx 0.3111$, matching the LP optimum. **Proposition 2.9 (explicit dual witness for Theorem 2.8).** In the dual of Proposition 2.5, write $q_t := \alpha_t - \beta_t$ so that $|q_t| \leq \gamma$ is enforced by $\alpha_t + \beta_t \leq \gamma$ and $\alpha_t, \beta_t \geq 0$. For the family in Theorem 2.8, the following choices are dual-feasible and attain the primal value:

$$\gamma := \frac{1}{4}, \quad q_1 := \frac{1}{4}, \quad q_t := -\frac{1}{4} \text{ for all } t \in \{2, \dots, n\}.$$

Let $y \in \mathbb{R}^n$ be

$$y_1 := \frac{3p}{4}, \quad y_2 := \frac{5(p-\tau)}{4}, \quad y_n := n\left(1 - p + \frac{\tau}{2}\right) + \frac{\tau}{4}, \quad y_s := 0 \text{ for } s \notin \{1, 2, n\}.$$

Setting $\alpha_t := \max\{q_t, 0\}$ and $\beta_t := \max\{-q_t, 0\}$ yields $\alpha_t, \beta_t \geq 0$ and $\alpha_t + \beta_t = |q_t| \leq \gamma$. One can verify that the dual constraints $y_s - \delta_s q_t \leq w_s t$ hold for every $1 \leq s \leq t \leq n$, with equality on the pairs used by the primal kernel of Theorem 2.8 (namely $(1, 1), (1, 2), (2, 2), (n, n)$). Finally, the dual objective equals the primal value:

$$\sum_s y_s - 2\tau\gamma = \text{OPT}_{\text{kern}}.$$

By strong duality, this certifies global optimality of the closed-form kernel. A complete mechanical check of all dual inequalities is performed by the script `stop_time_dual_cert_v18.py`.

Figure 2 (linear gap illustration). For the closed-form family in Theorem 2.8, the mixture suboptimality gap is

$$\text{OPT}_{\text{mix}} - \text{OPT}_{\text{kern}} = \frac{(n-2)\tau}{2} \left(1 - \frac{\tau}{p}\right),$$

which grows linearly in the horizon n for fixed τ, p . The plot below shows the gap for $\tau = 0.2$, $p = 0.9$ as a function of n .

Text-only replacement for Fig. 2. The plot is omitted (text-only submission). The separation family in Theorem 2.8 gives

$$\text{OPT}_{\text{mix}} - \text{OPT}_{\text{kern}} = \frac{(n-2)\tau}{2} \left(1 - \frac{\tau}{p}\right),$$

so the gap is linear in n for fixed (p, τ) . (To regenerate the original plot, run `make_figures_v67.py`.)

4.5 2.8 Support complexity beyond single-row randomization (new)

Theorem 2.8 shows that a single randomized row can already separate mixtures from optimal kernels. Empirically, however, optimal kernels can randomize in many rows. **Example 2.10 (two randomized rows, $n=8$)**. For $n = 8$ and $\tau = 0.2$, the LP optimum for

$$\begin{aligned} S_0 &\approx (0.339996, 0.069983, 0.000053, 0.019840, 0.055718, 0.071428, 0.001461, 0.441521), \\ S_1 &\approx (0.005131, 0.006532, 0.176472, 0.026986, 0.095191, 0.096904, 0.563583, 0.029202) \end{aligned}$$

randomizes in exactly two rows, $s \in \{1, 7\}$. In particular, the optimal row 7 splits between output times 7 and 8. The extracted dual variables satisfy the dual constraints and match the primal objective to numerical precision (primal-dual gap $< 10^{-15}$). **Example 2.11 (many randomized rows, $n=20$)**. For $n = 20$ and $\tau = 0.2$, an LP search over random Dirichlet instances found a case where the optimal kernel randomizes in 10 different rows while still achieving a mixture gap of about 0.60 in average expected stop time. This suggests that while Kernel-TV-Pad admits sparse optima (Proposition 2.4), the number of randomized rows can still grow with n . A structural characterization of when many rows must randomize is an open problem.

4.6 2.9 Multi-profile stop-time padding: pairwise vs star constraints (new)

So far we presented stop-time padding for two worlds (a binary secret). For Option B, it is useful to generalize to a K -ary secret $C \in [K]$ with completion-time laws S_c and padded stop-time laws $P_c = \Phi^\top S_c$ for a kernel-feasible padding kernel $\Phi(t \mid s)$. A natural *pairwise* StopTime-IND $_\tau$ constraint is:

$$\text{TV}(P_c, P_{c'}) \leq \tau \quad \forall c \neq c'.$$

Together with feasibility ($\Phi(t \mid s) = 0$ for $t < s$, $\sum_t \Phi(t \mid s) = 1$, $\Phi \geq 0$), minimizing average overhead $\frac{1}{K} \sum_c \mathbb{E}_{P_c}[T]$ remains a linear program by introducing standard absolute-value slack variables for each pair (c, c') . A simpler and often sufficient *star* formulation introduces a common reference distribution R and enforces

$$\text{TV}(P_c, R) \leq \tau/2 \quad \forall c,$$

which implies the pairwise constraints by the triangle inequality. The star LP uses only $O(Kn)$ TV constraints rather than $O(K^2n)$, at the cost of potential conservatism. This star form also connects directly to the posterior calculus in Section 7: if each P_c is additively close to a common R , then the stop time can be appended as one more step in Theorem 7.2 (see Section 6.3). **Proposition 2.12 (star can be strictly suboptimal, even when feasible)**. Consider $K = 3$ profiles, horizon $n = 5$, and privacy budget $\tau = 0.2$. Let success-time laws be

$$\begin{aligned} S_0 &\approx (0.0005, 0.6774, 0.2853, 0.0368, 0), \\ S_1 &\approx (0.2474, 0.1222, 0.6093, 0.0001, 0.0210), \\ S_2 &\approx (0.2749, 0.3145, 0.0529, 0.0674, 0.2903). \end{aligned}$$

Let the objective be the uniform-prior average expected stop time $\frac{1}{3} \sum_{c=0}^2 \mathbb{E}_{P_c}[T]$. (a) Under the *pairwise* constraints $\text{TV}(P_c, P_{c'}) \leq \tau$ for all $c \neq c'$, the LP optimum is $\text{OPT}_{\text{pair}} \approx 3.08872$, achieved by the kernel

$$\Phi_{\text{pair}} \approx \begin{pmatrix} 0.039647 & 0.960353 & 0 & 0 & 0 \\ 0 & 0.787304 & 0.212696 & 0 & 0 \\ 0 & 0 & 0.432964 & 0.120956 & 0.446079 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

(b) Under the *star* constraints $\text{TV}(P_c, R) \leq \tau/2$ for a common reference R , the LP optimum is $\text{OPT}_{\text{star}} \approx 3.22584$, achieved by

$$\Phi_{\text{star}} \approx \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0.624820 & 0.262765 & 0.005617 & 0.106798 \\ 0 & 0 & 0.450269 & 0.122898 & 0.426833 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad R \approx (0, 0.423753, 0.206459, 0.075668, 0.294120).$$

Therefore the star formulation is strictly conservative here: $\text{OPT}_{\text{star}} - \text{OPT}_{\text{pair}} \approx 0.13711$. *Verification.* The script `stop_time_star_gap_v21.py` solves both LPs and prints the optimal kernels and induced TVs; `stop_time_kary_cert_v38.py` additionally extracts HiGHS dual multipliers and checks KKT. **Certification remark.** Since both the pairwise and star formulations are linear programs, their optima admit short primal-dual certificates. The script `stop_time_kary_cert_v38.py` reruns the instance of Proposition 2.12, extracts the HiGHS dual multipliers, and checks the KKT conditions (primal feasibility, reduced-cost nonnegativity, complementary slackness), reporting a primal-dual gap on the order of 10^{-15} . This makes the numerical separation fully machine-checkable. Thus, while the star formulation reduces the number of TV constraints from $O(K^2n)$ to $O(Kn)$, it can be strictly suboptimal. We leave a tight characterization of when the star constraints are lossless (e.g., under geometric or convexity conditions on the feasible $\{P_c\}$) as an open problem.

Lemma 2.13 (TV diameter vs radius; barycentric star bound is optimal). Let $P_1, \dots, P_K$ be distributions on a finite alphabet, and let

$$D := \max_{i,j \in [K]} \text{TV}(P_i, P_j)$$

be their **diameter** in total variation. Define the **barycenter** $R := \frac{1}{K} \sum_{j=1}^K P_j$. Then for every $i \in [K]$,

$$\text{TV}(P_i, R) \leq \frac{K-1}{K} D.$$

Moreover, the constant $(K-1)/K$ is best possible in general: for the K point-masses $P_i = \delta_i$ on alphabet $\{1, \dots, K\}$, we have $D = 1$ but $\min_R \max_i \text{TV}(P_i, R) = 1 - 1/K = (K-1)/K$. *Proof.* By convexity of TV in its second argument,

$$\text{TV}(P_i, R) = \text{TV}\left(P_i, \frac{1}{K} P_i + \frac{1}{K} \sum_{j \neq i} P_j\right) \leq \frac{1}{K} \sum_{j \neq i} \text{TV}(P_i, P_j) \leq \frac{K-1}{K} D.$$

Tightness follows by direct calculation for the simplex vertices. $\square$

Figure 4 (TV-star relaxation across τ). For the fixed $K = 3, n = 5$ instance of Proposition 2.12, we sweep $\tau \in [0, 0.5]$ and solve both LPs. The TV-star optimum remains above the pairwise-TV optimum across an interval, illustrating *persistent* conservatism (not just a single tuned parameter point). The figure is generated by `make_figures_v67.py`.

Text-only replacement for Fig. 4. The plot is omitted (text-only submission). It swept $\tau \in [0, 0.5]$ for the $K=3, n=5$ instance in Proposition 2.12, showing a persistent interval where the TV-star relaxation stays above the pairwise-TV optimum. (To regenerate the original plot, run `make_figures_v67.py`.)

Discussion (why the TV-star shortcut can cost overhead). The TV-star constraints $\text{TV}(P_c, R) \leq \tau/2$ imply pairwise $\text{TV} \leq \tau$ by the triangle inequality. However, for $K \geq 3$ they can be much stronger than the pairwise constraints: if a family has pairwise diameter τ , then Lemma 2.13 guarantees the existence of a common center only within radius $(K-1)\tau/K$ (not $\tau/2$), and the worst-case gap factor

$$\frac{(K-1)\tau/K}{\tau/2} = 2 - 2/K$$

approaches 2 as K grows. In Section 10 we therefore treat star-style constraints as a **typing rule** enabling clean posterior composition, and we quantify the overhead cost separately (Proposition 2.12).

4.7 2.10 Posterior-friendly stop-time padding: L_∞ -star constraints and probability floors (new)

Section 6.3 composes the stop time with the sequential posterior calculus (Theorem 7.2) by treating the stop time as one more observation step. That calculus requires a **coordinate-wise** additive deviation bound relative to a secret-independent reference with a **positive lower bound**. This motivates a stop-time padding variant whose constraints are tailored to posterior control. **Problem 2.14 (PosteriorStopTime-Pad).** Fix a horizon $\{1, \dots, n\}$, success-time laws $\{S_c\}_{c \in [K]}$, and parameters (κ_T, δ_T) with $0 < \kappa_T < 1$ and $0 \leq \delta_T < \kappa_T$. Choose a kernel-feasible padding matrix Φ and a reference distribution R on $\{1, \dots, n\}$ to minimize average overhead

$$\min_{\Phi, R} \frac{1}{K} \sum_{c=1}^K \mathbb{E}_{P_c}[T] \quad \text{s.t.} \quad P_c = S_c \Phi, \quad \Phi \text{ kernel-feasible}, \quad \sum_t R(t) = 1, \quad R(t) \geq \kappa_T \quad \forall t,$$

$$|P_c(t) - R(t)| \leq \delta_T \quad \forall c, t.$$

This is a linear program: the constraints $|P_c(t) - R(t)| \leq \delta_T$ are two linear inequalities per (c, t) , and $R(t) \geq \kappa_T$ is linear. **Proposition 2.15 (PosteriorStopTime-Pad plugs into Theorem 7.2).** If (Φ, R) is feasible for Problem 2.14, then the padded stop time can be appended as an extra observation step with $\kappa_{T+1} = \kappa_T$ and $\delta_{T+1} = \delta_T$ in Theorem 7.2, yielding the posterior inflation factor $\log((\kappa_T + \delta_T)/(\kappa_T - \delta_T))$ contributed by the stop time. *Remark (certification).* Problem 2.14 is a linear program with a standard primal–dual optimum certificate. In particular, solvers such as SciPy/HiGHS return inequality marginals and equality marginals that can be checked against the KKT conditions (dual feasibility, complementary slackness, and a small primal–dual gap). For a small hard-coded instance, the script `stop_time_posterior_cert_v46.py` builds PosteriorStopTime-Pad, solves it, extracts HiGHS dual multipliers, and reports max KKT residuals on the order of 10^{-16} . *Remark (relation to TV-star constraints).* If one uses a TV-star constraint $\text{TV}(P_c, R) \leq \tau/2$, then $\|P_c - R\|_1 \leq \tau$ and hence $\|P_c - R\|_\infty \leq \tau$, so Problem 2.14 holds with $\delta_T = \tau$ provided one also enforces a floor $R(t) \geq \kappa_T$. However, $\|\cdot\|_\infty$ control can be much tighter than what TV alone implies, and we found concrete instances where TV-star constraints are strictly conservative relative to pairwise TV (Proposition 2.12).

5 3. Mark leakage: collapsing shard marks by public scheduling (PSC)

PSC fixes a public mapping from slot index s to a shard subset J_s and requires fixed-size traffic to every shard in J_s each slot (real or dummy). Then “which shard(s) were contacted” becomes a deterministic function of (epoch, s) . PSC collapses shard-level marks: conditioned on the public schedule, the set of contacted shards is a deterministic function of time, so only the within-set choice remains to be hidden. **Remark 3.1 (support containment and finite KL).** PSC restricts each slot to a public scheduled action set. If the cover rule assigns strictly positive probability to every action in the scheduled set, then every secret-dependent per-slot law has support contained in the cover support. This prevents the pathological case $D_{\text{KL}}(P\|Q) = \infty$ from support mismatch and makes the KL- and MI-based accounting in Sections 5–6 well-defined. —

6 4. Equalizing within a scheduled action set: MC-EQ gives a TV knob

6.1 4.1 TV shrinkage under mixture equalization

Let Q be a cover distribution and R a client’s true-target distribution over the same scheduled action set. If one lookup selects one action from R and $w - 1$ actions from Q , then the induced per-lookup contact distribution is

$$P = (1/w) R + (1 - 1/w) Q,$$

and

$$\text{TV}(P, Q) = \frac{1}{w} \text{TV}(R, Q).$$

Corollary 4.1 (history-uniform TV from one-real-among-w mixing). Let $K(x, \cdot)$ be a reference “cover” distribution (or a Markov kernel row) that does not depend on the secret profile C . Suppose the realized per-step action distribution conditioned on an arbitrary history h_{t-1} and secret $C = c$ has the mixture form

$$P_{c, h_{t-1}}(\cdot) = (1 - \rho) K(x, \cdot) + \rho R_{c, h_{t-1}}(\cdot)$$

for some $\rho \in [0, 1]$, some (possibly history- and secret-dependent) distribution $R_{c, h_{t-1}}$, and where x is the public state used to index the cover row (often x is simply the previous action). Then for every c and every history h_{t-1} ,

$$\text{TV}(P_{c, h_{t-1}}, K(x, \cdot)) = \rho \text{TV}(R_{c, h_{t-1}}, K(x, \cdot)) \leq \rho.$$

In particular, in the common $\rho = 1/w$ “one real among w ” regime, the **history-uniform TV deviation** parameter satisfies $\delta_{eq} \leq 1/w$ *automatically*, without any additional assumptions on how $R_{c, h}$ depends on the past. *Proof.* Immediate from the convexity identity $\text{TV}((1 - \rho)Q + \rho R, Q) = \rho \text{TV}(R, Q) \leq \rho$. **Lemma 4.2 (pointwise deviation under one-real-among-w mixing).** Let K be a reference distribution on a finite set J , let R be any distribution on J , and define

$$P := (1 - \rho)K + \rho R.$$

Then for every $y \in J$,

$$|P(y) - K(y)| = \rho |R(y) - K(y)| \leq \rho.$$

If additionally K is uniform on $|J| = m$, then $\|P - K\|_\infty \leq \rho(1 - 1/m) < \rho$. In the common $\rho = 1/w$ regime, this provides a per-step additive deviation parameter $\delta_\infty \leq 1/w$ for Theorem 7.2 and Theorem 10.1. *Proof.* Immediate from $P - K = \rho(R - K)$ and $|R(y) - K(y)| \leq 1$ (or $\leq 1 - 1/m$ in the uniform case). —

7 5. New: exact and moment KL bounds for “one-real-among- w ” injection

7.1 5.1 Exact per-step KL for injection into a reference distribution

Let q be a reference distribution on a finite set A with $q_j > 0$. Fix $\rho = 1/w$, and define

$$p_j := (1 - \rho)q + \rho\delta_j.$$

Proposition 5.1 (exact KL). For every j ,

$$D_{\text{KL}}(p_j \| q) = ((1 - \rho)q_j + \rho) \log\left(1 - \rho + \frac{\rho}{q_j}\right) + (1 - \rho)(1 - q_j) \log(1 - \rho).$$

Proof. Split the KL sum into coordinate j and $A \setminus \{j\}$; for $y \neq j$, $p_j(y) = (1 - \rho)q(y)$, giving the common $\log(1 - \rho)$ term. **Uniform special case.** If q is uniform on $|A| = m$, then

$$D_{\text{KL}}(p_j \| q) = \frac{1 + \rho(m - 1)}{m} \log(1 + \rho(m - 1)) + \frac{(m - 1)(1 - \rho)}{m} \log(1 - \rho) = \frac{m - 1}{2w^2} + O\left(\frac{m^2}{w^3}\right).$$

7.2 5.2 A moment upper bound avoiding a global $q_{\min}$

Lemma 5.2 (moment bound). For $p_j = (1 - \rho)q + \rho\delta_j$,

$$D_{\text{KL}}(p_j \| q) \leq \rho^2 \left(\frac{1}{q_j} - 1 \right).$$

Proof. Use $\log(1 + x) \leq x$ for $x \geq 0$ and $\log(1 - \rho) \leq -\rho$; the linear-in- ρ terms cancel. **Corollary 5.3.** If the injected index J is random, then

$$\mathbb{E}[D_{\text{KL}}(p_J \| q)] \leq \rho^2 \left(\mathbb{E}\left[\frac{1}{q_J}\right] - 1 \right).$$

This suggests designing schedules/covers so that “real” contacts land on higher-mass actions under q . —

8 6. Dependency-robust MI accounting and Fano inversion

The posterior-first path (Theorem 10.1) gives **transcript-by-transcript** guarantees but can be conservative over long horizons. This section provides a complementary **average-information** tool: an MI upper bound that remains valid even when the real process depends arbitrarily on the full history, as long as each step is TV-close to a secret-independent Markov reference kernel.

8.1 6.1 Markov-reference MI bound

Let C be a hidden profile and let $O_{1:T}$ be the observed mark process (after PSC has reduced marks to within-set choices). For each c , write P_c for the law of $O_{1:T}$ conditioned on $C = c$; P_c may be non-Markov and may depend on the full history.

Fix a **reference** Markov chain Q on the mark space $\mathcal{X}$ with initial distribution μ and kernel K :

$$Q(o_{1:T}) = \mu(o_1) \prod_{t=2}^T K(o_{t-1}, o_t).$$

Assume a positive floor

$$k_{\min} := \min_{x,y \in \mathcal{X}} K(x,y) > 0.$$

Assume the following **history-uniform TV deviation** contract holds for the real process: for every $t \geq 2$, every profile c , every history h_{t-1} whose last state is x , and every measurable set $A \subseteq \mathcal{X}$,

$$\text{TV}(P(O_t \in \cdot \mid C = c, H_{t-1} = h_{t-1}), K(x, \cdot)) \leq \delta_{eq}. \quad (\text{HUTV})$$

(Under one-real-among- w MC-EQ, Corollary 4.1 gives $\delta_{eq} \leq 1/w$ automatically.)

Theorem 6.1 (MI via a Markov reference). Under (HUTV) with $\delta_{eq} < k_{\min}$,

$$I(C; O_{1:T}) \leq I(C; O_1) + (T-1) \ln \left(1 + \frac{2\delta_{eq}^2}{k_{\min}} \right).$$

More generally, for any choice of reference initial law μ ,

$$I(C; O_{1:T}) \leq \mathbb{E}_c[D_{\text{KL}}(P_c(O_1) \parallel \mu)] + (T-1) \ln \left(1 + \frac{2\delta_{eq}^2}{k_{\min}} \right).$$

Proof. Fix any reference distribution Q that does not depend on C . A standard variational bound gives

$$I(C; O_{1:T}) = \mathbb{E}_c[D_{\text{KL}}(P_c \parallel P)] \leq \mathbb{E}_c[D_{\text{KL}}(P_c \parallel Q)],$$

where P is the unconditional law of $O_{1:T}$ and we used that $D_{\text{KL}}(\cdot \parallel \cdot)$ is minimized at P .

Now apply the chain rule for KL:

$$D_{\text{KL}}(P_c \parallel Q) = D_{\text{KL}}(P_c(O_1) \parallel \mu) + \sum_{t=2}^T \mathbb{E}_{h_{t-1} \sim P_c} [D_{\text{KL}}(P(O_t \mid c, h_{t-1}) \parallel K(x, \cdot))],$$

where x is the last state of h_{t-1} .

For each conditional term, (HUTV) gives $\text{TV} \leq \delta_{eq}$ and the reference row has minimum mass $\geq k_{\min}$, so Lemma 10.3 yields

$$D_{\text{KL}}(P(O_t \mid c, h_{t-1}) \parallel K(x, \cdot)) \leq \ln \left(1 + \frac{2\delta_{eq}^2}{k_{\min}} \right),$$

uniformly over c and h_{t-1} . Summing over $t = 2, \dots, T$ and taking $\mathbb{E}_c$ gives the second displayed bound.

To obtain the first bound, choose $\mu := P(O_1)$; then $\mathbb{E}_c[D_{\text{KL}}(P_c(O_1) \parallel P(O_1))] = I(C; O_1)$. $\square$

8.2 6.2 Fano inversion and an explicit w rule

If $C \sim \text{Unif}([K])$ and $I(C; O_{1:T}) \leq B$, then Fano's inequality implies

$$P_{\text{succ}} := \Pr[\hat{C}(O_{1:T}) = C] \leq \frac{B + \ln 2}{\ln K}.$$

Combining this with Theorem 6.1 and the MC-EQ plug-in $\delta_{eq} \leq 1/w$ yields the explicit sufficient condition

$$\frac{(T-1) \ln \left(1 + \frac{2}{k_{\min} w^2}\right) + I(C; O_1) + \ln 2}{\ln K} \leq \alpha \quad \Rightarrow \quad P_{\text{succ}} \leq \alpha,$$

which can be inverted for w given a target success probability α . For small $1/(k_{\min} w^2)$ this behaves like

$$w \gtrsim \sqrt{\frac{2(T-1)}{k_{\min} (\alpha \ln K - \ln 2 - I(C; O_1))}}.$$

In contrast to posterior-first composition (Theorem 10.1), this MI/Fano path scales roughly like $\sqrt{T}$ in the high- w regime. —

8.3 6.3 End-to-end compiler calculus: folding stop time into the same posterior calculus (new)

Option B's goal is to reason about *joint* transcripts that include both (i) a contact-mark trace and (ii) a termination time. A convenient way to make these modules compose is to treat the stop time as just **one more observation step** and apply the sequential posterior bound from Theorem 7.2. Let $C \sim \text{Unif}([K])$. Consider an observation sequence

$$(O_1, \dots, O_T, \tilde{T}),$$

where $O_{1:T}$ is a contact-mark trace and $\tilde{T}$ is the padded stop time produced by a stop-time compiler. Assume: 1. (**Mark compiler hypothesis**) For each step $t \leq T$ there is a secret-independent reference $K_t(\cdot | h)$ with $\kappa_t := \inf_{h,y} K_t(y | h) > 0$ and an additive deviation bound

$$|P(O_t = y | C = c, H_{t-1} = h) - K_t(y | h)| \leq \delta_t < \kappa_t.$$

2. (**Stop-time compiler hypothesis**) There is a secret-independent reference Q on stop times with $\kappa_{T+1} := \min_y Q(y) > 0$ such that for all c and all histories h ,

$$|P(\tilde{T} = y | C = c, H_T = h) - Q(y)| \leq \delta_{T+1} < \kappa_{T+1}.$$

Then Theorem 7.2 applies to the length- $T+1$ transcript and yields, for every realized transcript,

$$\max_c P(C = c | O_{1:T}, \tilde{T}) \leq \exp(\varepsilon_{T+1})/K, \quad \varepsilon_{T+1} = \sum_{t=1}^{T+1} \log \left(\frac{\kappa_t + \delta_t}{\kappa_t - \delta_t} \right).$$

Concrete plug-ins. - Under PSC with uniform scheduled sets $|J_t| = m_t$, the one-real-among- w mixing form together with Lemma 4.2 gives the pointwise deviation bound $\delta_t \leq 1/w$. Consequently $\varepsilon_T \approx \sum_t 2m_t/w$ when m_t/w is small. - Under a *mixture* stop-time compiler that outputs $\tilde{T} \sim Q$ with probability $1 - \lambda$ and otherwise outputs an arbitrary $\tilde{T}$ (subject to feasibility), one has the additive deviation bound $\delta_{T+1} \leq \lambda$ for every y because $P(\cdot) - Q(\cdot) = \lambda(R(\cdot) - Q(\cdot))$ with $|R(y) - Q(y)| \leq 1$. This yields an explicit end-to-end posterior control inequality coupling the two knobs w (mark equalization) and λ (stop-time mixing). In particular, a sufficient condition for $\max_c P(C = c | \text{transcript}) \leq (1 + \eta)/K$ is $\varepsilon_{T+1} \leq \log(1 + \eta)$, i.e. a sum of one term from each step.

9 7. New: near-1/K identification via pointwise maximal leakage from TV-row perturbation

9.1 7.1 PML definition and a posterior bound under a uniform prior

For a channel $W_u(y | c)$ (at fixed state u), define the pointwise maximal leakage under a uniform prior P_C as

$$\text{PML}_u := \sup_y \log \frac{\max_c W_u(y | c)}{\sum_c P_C(c) W_u(y | c)}.$$

This is a pointwise (per-output) variant of **maximal leakage** [IWM18]. The following identity makes it directly interpretable as a posterior bound: Under a uniform prior, $\max_c \Pr[C = c | y] = e^{\text{PML}_u} / K$. Hence bounding PML_u gives a sharp near-1/K identification guarantee.

9.2 7.2 Additive perturbation $\Rightarrow$ finite PML (and thus near-1/K posterior control)

Assume a reference row $K(u, \cdot)$ has a uniform lower bound $\kappa := \min_y K(u, y)$, and assume an additive deviation bound $\delta_\infty := \sup_{c,y} |W_u(y | c) - K(u, y)|$. If $\kappa > 2\delta_\infty$, then

$$\text{PML}_u \leq \sup_y \log \frac{K(u, y) + \delta_\infty}{K(u, y) - \delta_\infty} \leq \log \frac{\kappa + \delta_\infty}{\kappa - \delta_\infty}.$$

Moreover, in general $\delta_\infty \leq 2 \text{TV}(W_u(\cdot | c), K(u, \cdot))$. Under the concrete one-real-among- w mixing form $W_u = (1 - 1/w)K(u, \cdot) + (1/w)R$, Lemma 4.2 sharpens this to $\delta_\infty \leq 1/w$. Thus for $\kappa = \Theta(1/d)$ (lazy bounded-degree cover), $\text{PML}_u \approx O(d/w)$. **Corollary 7.1 (explicit w for near-1/K posterior).** Fix $\eta \in (0, 1)$ and target $\max_c \Pr[C = c | y] \leq (1 + \eta)/K$ (i.e., within a factor $1 + \eta$ of random guessing). It suffices that $\text{PML}_u \leq \ln(1 + \eta)$, i.e.

$$\frac{\kappa + \delta_\infty}{\kappa - \delta_\infty} \leq 1 + \eta \iff \delta_\infty \leq \frac{\eta}{2 + \eta} \kappa.$$

Under the plug-in $\delta_\infty \leq 1/w$ (Lemma 4.2), a sufficient condition is

$$w \geq \frac{(2 + \eta)}{\eta \kappa}.$$

In particular, for a lazy bounded-degree cover kernel with $\kappa \geq 1/(2d)$, it suffices that

$$w \geq \frac{2d(2 + \eta)}{\eta} \approx \frac{4d}{\eta}.$$

Remark. This route is designed for the regime where one wants guarantees close to random guessing $1/K$; MI $\rightarrow$ Fano can be vacuous there. —

9.3 7.3 New: sequential posterior composition under per-history additive deviation

The per-step PML bound above is local (one step / one state). For a full transcript $y_{1:T}$ we can control the posterior directly under a mild **per-history additive deviation** hypothesis. **Theorem 7.2 (Sequential posterior bound from per-history additive deviation).** Assume C is uniform on $[K]$. For each time t and each history h_{t-1} , let $K_t(\cdot | h_{t-1})$ be a reference distribution

that does not depend on C , and define $\kappa_t := \inf_{h,y} K_t(y | h)$. Suppose the real conditional laws satisfy, for all secrets c , histories h , and outcomes y , $|P(O_t = y | C = c, H_{t-1} = h) - K_t(y | h)| \leq \delta_t$, with $0 \leq \delta_t < \kappa_t$. Then for every transcript $y_{1:T}$, $\max_c P(C = c | O_{1:T} = y_{1:T}) \leq \exp(\epsilon_T)/K$, where $\epsilon_T := \sum_{t=1}^T \log((\kappa_t + \delta_t)/(\kappa_t - \delta_t))$. *Proof.* For any c, c' and any history h , the hypothesis implies $P(O_t = y | c, h) \leq K_t(y | h) + \delta_t$ and $P(O_t = y | c', h) \geq K_t(y | h) - \delta_t$, so the conditional likelihood ratio is bounded by $(\kappa_t + \delta_t)/(\kappa_t - \delta_t)$. Multiply these bounds over time (chain rule for likelihoods) to bound the transcript likelihood ratio. With a uniform prior, Bayes' rule implies the stated posterior maximum. **Corollary 7.3 (PSC + MC-EQ plug-in; linear w budget).** If K_t is uniform on a scheduled set J_t of size m_t , then $\kappa_t = 1/m_t$. If one only knows a per-history TV deviation bound $\text{TV}(P(\cdot | c, h), K_t(\cdot | h)) \leq \delta_{eq,t}$, then the pointwise deviation satisfies the generic inequality

$$\delta_t := \sup_{c,h,y} |P(O_t = y | c, h) - K_t(y | h)| \leq 2\delta_{eq,t}.$$

In the common one-real-among- w mixing form on J_t ,

$$P(\cdot | c, h) = (1 - 1/w)K_t(\cdot | h) + (1/w)R_{c,h}(\cdot),$$

Lemma 4.2 yields the sharper bound $\delta_t \leq 1/w$. Provided $w > m_t$ for all t , Theorem 7.2 gives

$$\epsilon_T \leq \sum_{t=1}^T \log\left(\frac{1/m_t + 1/w}{1/m_t - 1/w}\right).$$

For small m_t/w , this is approximately $\sum_t 2m_t/w$. Therefore, to ensure $\max_c P(C = c | y) \leq (1 + \epsilon_T)/K$, it suffices to have $\epsilon_T \leq \log(1 + \epsilon_T)$, giving the rule

$$w \gtrsim (2/\log(1 + \epsilon_T)) \sum_{t=1}^T m_t \approx (2/\epsilon_T) \sum_{t=1}^T m_t.$$

Remark. This is a worst-case composition bound. In long-horizon regimes, MI/KL accounting can scale like $\sqrt{T}$ for the same primitive, while Theorem 7.2 scales linearly in the number of perturbed steps. Theorem 7.2 is most useful for short sessions, or when one can bound the number of “real injections” explicitly.

10 8. Binary distinguishing: robust $\text{KL} \rightarrow \text{TV}$ envelopes and non-vacuous design rules

To upper bound $\text{TV}(P, Q)$ from a KL bound $D := D_{\text{KL}}(P || Q)$, record Pinsker and Bretagnolle–Huber and the robust envelope

$$g(D) := \min\{\sqrt{D/2}, \sqrt{1 - e^{-D}}\}, \quad \text{TV}(P, Q) \leq g(D).$$

Consequently, any binary distinguisher between two transcript laws P, Q with $D_{\text{KL}}(P || Q) \leq D$ has advantage at most $g(D)$. In particular, to ensure $\text{TV}(P, Q) \leq \epsilon$, it suffices that

$$D \leq 2\epsilon^2 \quad (\text{Pinsker}) \quad \text{or} \quad D \leq -\ln(1 - \epsilon^2) \quad (\text{Bretagnolle–Huber}).$$

The Bretagnolle–Huber inversion remains informative for moderate ϵ where the small- D Pinsker regime can be overly strict. —

11 9. Mixing-time stability under equalization

Section 6 bounds *privacy* leakage of the observation sequence relative to a Markov reference. Here we bound a different failure mode: equalization perturbs the intended cover process, potentially destroying the *routing/latency* benefits (mixing and stationary behavior) of the cover Markov chain. We model the scheduled cover process as a Markov kernel K on a finite state space X (e.g., committees). The realized process uses a kernel K_c (possibly secret-dependent) that is row-wise close to K . **Assumption (row-wise TV perturbation).** Let $\delta_{eq} := \sup_x \text{TV}(K_c(x, \cdot), K(x, \cdot))$. In MC-EQ-type mechanisms, δ_{eq} typically scales like $\Theta(1/w)$.

11.1 9.1 One-step contractive case ($\beta < 1$)

Define the Dobrushin (TV contraction) coefficient $\beta(P) := \sup_{\{x, x'\}} \text{TV}(P(x, \cdot), P(x', \cdot))$. Then for any distributions μ, ν , we have $\|\mu P - \nu P\|_{\text{TV}} \leq \beta(P) \|\mu - \nu\|_{\text{TV}}$. **Lemma 9.1 (geometric perturbation recursion).** Let $\Delta_t := \|\mu K_c^t - \mu K^t\|_{\text{TV}}$. If $\beta(K) = \beta < 1$, then $\Delta_{t+1} \leq \delta_{eq} + \beta \Delta_t$, hence $\Delta_t \leq \delta_{eq} (1 - \beta^t) / (1 - \beta) \leq \delta_{eq} / (1 - \beta)$. Proof sketch. Add/subtract $(\mu K_c^t K)K$: $\Delta_{t+1} = \|\mu K_c^t K_c - \mu K^t K\| \leq \|\mu K_c^t (K_c - K)\| + \|(\mu K_c^t K) - (\mu K^t K)\| \leq \delta_{eq} + \beta \Delta_t$. Unroll the recursion. $\square$

11.2 9.2 Block contraction for sparse kernels and the $w \gtrsim t_{\text{mix}}$ rule

For sparse local kernels, one can have $\beta(K) = 1$ because different rows can have disjoint supports, making Lemma 9.1 inapplicable. One remedy is a block coefficient $\tau_m := \beta(K^m)$, which becomes < 1 once m exceeds the chain's mixing scale. **Lemma 9.2 (block-time geometric bound).** Let $\delta_m := \sup_x \text{TV}(K_c^m(x, \cdot), K^m(x, \cdot))$. Then for any $k \geq 1$, $\|\mu K_c^{\{km\}} - \mu K^{\{km\}}\|_{\text{TV}} \leq \delta_m (1 - \tau_m^k) / (1 - \tau_m) \leq \delta_m / (1 - \tau_m)$. A crude but general bound is $\delta_m \leq m \delta_{eq}$ (telescoping), so the stability term is $O(m \delta_{eq} / (1 - \tau_m))$. To pick m , relate τ_m to the mixing profile $d_K(t) := \max_x \text{TV}(K^t(x, \cdot), \pi)$. **Lemma 9.3 (τ_m from mixing).** For all $t \geq 1$, $\beta(K^t) \leq 2 d_K(t)$. In particular, letting $m = t_{\text{mix}}(K, 1/4)$ (so $d_K(m) \leq 1/4$) gives $\tau_m \leq 1/2$. **Design consequence (mixing-time preservation).** In the MC-EQ regime $\delta_{eq} = \Theta(1/w)$, and taking $m \approx t_{\text{mix}}(K)$ so that τ_m is bounded away from 1, Lemma 9.2 suggests that keeping the realized process within $O(\varepsilon)$ TV of the cover process on the natural time scale $t \approx t_{\text{mix}}(K)$ typically requires $w \gtrsim t_{\text{mix}}(K) / \varepsilon$. This formalizes the heuristic that w is simultaneously a privacy knob (Sections 6-7) and a stability knob for the cover process itself.

12 10. Flagship compiler calculus theorem: composing PSC, MC-EQ, and stop-time padding

We now state a single “plug-and-play” theorem that composes the main compiler modules in this draft. It has two variants: (i) a **posterior control** variant (near- $1/K$ guarantees) based on Theorem 7.2, and (ii) an **MI/Fano** variant (high-confidence identification control) based on the Markov-reference theorem.

12.1 10.1 Posterior-control variant (near- $1/K$)

Theorem 10.1 (Compiler calculus: sequential posterior control for joint traces). Let $C \sim \text{Unif}([K])$ be a hidden profile. Consider a joint transcript $(O_{1:T}, \tilde{T})$ consisting of a contact-mark trace $O_{1:T}$ and a padded stop time $\tilde{T}$. Assume: 1. (**PSC / public schedule**) The set of admissible

actions at time t is a public set J_t of size m_t , and the reference distribution $K_t(\cdot | h)$ supported on J_t satisfies a lower bound $\kappa_t := \inf_{h,y} K_t(y | h) > 0$. 2. (**MC-EQ / per-history additive deviation**) For each $t \leq T$, for all profiles c , histories h , and outcomes y ,

$$|P(O_t = y | C = c, H_{t-1} = h) - K_t(y | h)| \leq \delta_t < \kappa_t.$$

A sufficient condition is the one-real-among- w mixing form, in which Lemma 4.2 gives the pointwise deviation bound $\delta_t \leq 1/w$ (and Corollary 4.1 gives the TV deviation $\delta_{eq,t} \leq 1/w$). 3. (**Stop-time padding with a posterior-friendly reference**) There exists a secret-independent reference distribution R on $\{1, \dots, n\}$ with $\kappa_{T+1} := \min_t R(t) > 0$ such that for all c, h, y ,

$$|P(\tilde{T} = y | C = c, H_T = h) - R(y)| \leq \delta_{T+1} < \kappa_{T+1}.$$

For example, this holds if one solves PosteriorStopTime-Pad (Problem 2.14) with parameters (κ_T, δ_T) . Then for every realized transcript $(y_{1:T}, t)$,

$$\max_c \Pr[C = c | O_{1:T} = y_{1:T}, \tilde{T} = t] \leq \frac{\exp(\varepsilon_{T+1})}{K}, \quad \varepsilon_{T+1} := \sum_{t=1}^{T+1} \log\left(\frac{\kappa_t + \delta_t}{\kappa_t - \delta_t}\right).$$

In particular, to guarantee near-random identification $\max_c \Pr[C = c | \text{transcript}] \leq (1 + \eta)/K$, it suffices that $\varepsilon_{T+1} \leq \log(1 + \eta)$. *Proof.* This is exactly Theorem 7.2 applied to the length- $(T+1)$ transcript. Assumptions (1)-(3) are the per-history additive deviation hypotheses with a positive lower bound. **Design plug-in (uniform scheduled sets).** If K_t is uniform on J_t (so $\kappa_t = 1/m_t$) and one-real-among- w mixing gives $\delta_t \leq 1/w$ (more sharply, $\delta_t = (1 - 1/m_t)/w$ in the canonical injection), then

$$\varepsilon_{T+1} \leq \sum_{t=1}^T \log\left(\frac{1/m_t + 1/w}{1/m_t - 1/w}\right) + \log\left(\frac{\kappa_T + \delta_T}{\kappa_T - \delta_T}\right).$$

When m_t/w is small this is approximately $\sum_t 2m_t/w$ plus the stop-time term (using $\log \frac{1+a}{1-a} = 2a + O(a^3)$ with $a = m_t/w$).

12.2 10.2 MI/Fano variant (high-confidence identification)

Theorem 10.2 (Compiler calculus: MI bound + Fano inversion). Assume the Markov-reference hypothesis of Section 6 holds for $O_{1:T}$ with history-uniform TV deviation δ_{eq} and reference kernel minimum $k_{\min} > 0$. Assume also that the stop-time compiler satisfies a per-history TV deviation bound $\text{TV}(P(\tilde{T} | C = c, H_T = h), R(\cdot | h)) \leq \delta_T$ for some secret-independent reference $R(\cdot | h)$ with $\kappa_T := \inf_{h,t} R(t | h) > 0$. Then

$$I(C; O_{1:T}, \tilde{T}) \leq (T-1) \ln\left(1 + \frac{2\delta_{eq}^2}{k_{\min}}\right) + \ln\left(1 + \frac{2\delta_T^2}{\kappa_T}\right) + I(C; O_1).$$

Consequently, if C is uniform on $[K]$, Fano gives $P_{\text{succ}} \leq (I + \ln 2)/\ln K$. **Lemma 10.3 (KL from TV and a minimum mass).** Let P, Q be distributions on a finite set with $q_{\min} := \min_y Q(y) \geq \kappa > 0$ and $\text{TV}(P, Q) \leq \delta < \kappa$. Then

$$D_{\text{KL}}(P \| Q) \leq \ln\left(1 + \frac{2\delta^2}{\kappa}\right).$$

Proof. Let $\Delta(y) := P(y) - Q(y)$ and write $\Delta_+(y) := \max\{\Delta(y), 0\}$ and $\Delta_-(y) := \max\{-\Delta(y), 0\}$. Then $\sum_y \Delta_+(y) = \sum_y \Delta_-(y) = \text{TV}(P, Q) = \delta$ and

$$\sum_y \Delta(y)^2 = \sum_y \Delta_+(y)^2 + \sum_y \Delta_-(y)^2 \leq \delta^2 + \delta^2 = 2\delta^2.$$

Hence the chi-square divergence satisfies

$$\chi^2(P\|Q) := \sum_y \frac{(P(y) - Q(y))^2}{Q(y)} \leq \frac{1}{\kappa} \sum_y \Delta(y)^2 \leq \frac{2\delta^2}{\kappa}.$$

Finally, by Jensen's inequality,

$$D_{\text{KL}}(P\|Q) = \mathbb{E}_P[\ln(P/Q)] \leq \ln \mathbb{E}_P[P/Q] = \ln \sum_y \frac{P(y)^2}{Q(y)} = \ln(1 + \chi^2(P\|Q)),$$

which yields the claim. $\square$ *Proof of Theorem 10.2.* By the Markov-reference theorem (Section 6),

$$I(C; O_{1:T}) \leq (T-1) \ln \left(1 + \frac{2\delta_{eq}^2}{k_{\min}} \right) + I(C; O_1).$$

For the padded stop time, condition on the history $H_T = h$. Let $P_{c,h}$ denote the conditional law of $\tilde{T}$ given $(C = c, H_T = h)$, and let $\bar{P}_h := \frac{1}{K} \sum_c P_{c,h}$ be the profile mixture. Then

$$I(C; \tilde{T} \mid H_T = h) = \frac{1}{K} \sum_c D_{\text{KL}}(P_{c,h} \parallel \bar{P}_h) \leq \frac{1}{K} \sum_c D_{\text{KL}}(P_{c,h} \parallel R_h),$$

since $\sum_c D(P_{c,h} \parallel \bar{P}_h) = \sum_c D(P_{c,h} \parallel R_h) - K D(\bar{P}_h \parallel R_h) \leq \sum_c D(P_{c,h} \parallel R_h)$. By the assumed per-history TV bound and Lemma 10.3,

$$D_{\text{KL}}(P_{c,h} \parallel R_h) \leq \ln \left(1 + \frac{2\delta_T^2}{\kappa_T} \right)$$

for every c, h . Thus $I(C; \tilde{T} \mid H_T) \leq \ln(1 + 2\delta_T^2/\kappa_T)$. Finally, by the chain rule,

$$I(C; O_{1:T}, \tilde{T}) = I(C; O_{1:T}) + I(C; \tilde{T} \mid O_{1:T}) \leq I(C; O_{1:T}) + I(C; \tilde{T} \mid H_T),$$

which gives the stated bound. $\square$

12.3 10.3 A small sanity check (non-adversarial)

To gauge constant factors, we simulated the simplest i.i.d. model consistent with Theorem 10.1: $K = 200$ profiles, a uniform scheduled set of size $m = 20$, horizon $T = 20$, one-real-among- w injection with $w = 2000$, and a stop-time mixture with $n = 10$ and $\lambda = 0.05$. Across 2,000 trials, the empirical median of $\max_c \Pr[C = c \mid \text{transcript}]$ was about 0.00725, while the Theorem 7.2 bound predicted ≤ 0.0193 . Meanwhile, an MI/Fano accounting bound was numerically much looser in this near- $1/K$ regime. See `compiler_calculus_sims_v23.py`.

12.4 10.4 Design worksheet (worked example)

This subsection shows how to instantiate Theorem 10.1 with concrete numbers. We emphasize that the resulting bounds are *worst-case* and can be conservative; the goal is a simple, checkable knob-setting recipe. **Inputs.** Choose: - number of profiles K (uniform prior), - trace horizon T , - public scheduled set sizes $m_t := |J_t|$, - desired posterior slack η (target $\max_c \Pr[C = c \mid \text{transcript}] \leq (1 + \eta)/K$), - stop-time reference support size n and a stop-time additive deviation target $(\kappa_{T+1}, \delta_{T+1})$. **Step 1 (mark-trace term).** If each step uses one-real-among- w mixing inside a uniform scheduled set of size m_t and $w > m_t$, then $\kappa_t = 1/m_t$ and Lemma 4.2 gives a per-step pointwise deviation $\delta_t \leq 1/w$ (sharper: $\delta_t = (1 - 1/m_t)/w$ in the canonical injection). Compute

$$\varepsilon_{\text{trace}}(w) := \sum_{t=1}^T \ln \left(\frac{1/m_t + 1/w}{1/m_t - 1/w} \right).$$

Step 2 (stop-time term). Choose (or optimize for) a secret-independent stop-time reference R on $\{1, \dots, n\}$ with $\kappa_{T+1} := \min_t R(t)$. One can enforce the needed additive deviation and floor constraints by solving PosteriorStopTime-Pad (Problem 2.14), and then ensure $|P(\tilde{T} = t \mid c, h) - R(t)| \leq \delta_{T+1} < \kappa_{T+1}$ for all c, h, t . For example, if R is uniform then $\kappa_{T+1} = 1/n$, and a conservative sufficient condition under a simple mixture stop-time compiler is $\delta_{T+1} \leq \lambda$. A convenient way to enforce these pointwise and floor constraints is to solve **PosteriorStopTime-Pad** (Problem 2.14) for the desired $(\kappa_{T+1}, \delta_{T+1})$; any feasible padding kernel from that LP plugs directly into Theorem 10.1 via Proposition 2.15. **Step 3 (budgeting).** Theorem 10.1 yields

$$\max_c \Pr[C = c \mid \text{transcript}] \leq \frac{\exp(\varepsilon_{\text{trace}}(w) + \varepsilon_{\text{stop}})}{K}, \quad \varepsilon_{\text{stop}} := \ln \left(\frac{\kappa_{T+1} + \delta_{T+1}}{\kappa_{T+1} - \delta_{T+1}} \right).$$

Thus it suffices to pick w and stop-time parameters so that $\varepsilon_{\text{trace}}(w) + \varepsilon_{\text{stop}} \leq \ln(1 + \eta)$. **Worked examples.** Take $K = 10^5$, $T = 50$, and uniform scheduled sets of size $m_t \equiv 20$. For stop time, take $n = 30$ with uniform reference R , so $\kappa_{T+1} = 1/30$, and target $\delta_{T+1} = 0.001$, which gives $\varepsilon_{\text{stop}} = \ln((1/30 + 0.001)/(1/30 - 0.001)) \approx 0.0600$. (A) **Near-random guessing within a factor 2.** Aim for $\max_c \Pr[C = c \mid \text{transcript}] \leq 2/K$ (i.e. $\eta = 1$, so $\ln(1 + \eta) = \ln 2$). This leaves an allowable trace budget of about 0.6331. Solving $\varepsilon_{\text{trace}}(w) \leq 0.6331$ yields $w = 3159$ as the smallest integer meeting the bound:

$$\varepsilon_{\text{trace}}(3159) = 50 \ln \left(\frac{1/20 + 1/3159}{1/20 - 1/3159} \right) \approx 0.63312, \quad \varepsilon_{\text{trace}} + \varepsilon_{\text{stop}} \approx 0.69314 \leq \ln 2.$$

(B) **Within a factor 10.** Aim for $\max_c \Pr[C = c \mid \text{transcript}] \leq 10/K$ (i.e. $\eta = 9$, $\ln(1 + \eta) = \ln 10$). Then the smallest feasible w is $w = 892$:

$$\varepsilon_{\text{trace}}(892) \approx 2.24253, \quad \varepsilon_{\text{trace}} + \varepsilon_{\text{stop}} \approx 2.30253 \leq \ln 10.$$

(C) **Within a factor 100.** Aim for $\max_c \Pr[C = c \mid \text{transcript}] \leq 100/K$ (i.e. $\eta = 99$, $\ln(1 + \eta) = \ln 100$). Then the smallest feasible w is $w = 441$.

(D) **Non-uniform public schedules.** Consider $T = 100$ slots split into two public schedule sizes: 50 slots with $m = 8$ and 50 slots with $m = 32$. Take $n = 60$ and $\delta_{T+1} = 0.001$, so

$$\varepsilon_{\text{stop}} = \ln \left(\frac{1/60 + 0.001}{1/60 - 0.001} \right) \approx 0.12014.$$

For posterior targets (uniform prior over K profiles):

- **Within a factor 2.** For $\eta = 1$ (target $\leq 2/K$), the minimal feasible w under the loose per-step ratio is $w = 6981$.
- **Within a factor 10.** For $\eta = 9$ (target $\leq 10/K$), the minimal feasible w is $w = 1833$.
- **Within a factor 100.** For $\eta = 99$ (target $\leq 100/K$), the minimal feasible w is $w = 893$.

Under the canonical injection ratio (uniform cover on the scheduled set), these improve to $w = 3478, 904, 434$ respectively. All values are reproduced by `compiler_design_worksheet_latest.py --examples`.

These examples illustrate that **posterior-first** guarantees become expensive when the goal is extremely close to random guessing (small η). **Contrast (MI/Fano route).** If instead one only seeks a high-confidence identification bound, e.g. $P_{\text{succ}} \leq 0.2$ among $K = 10^5$ profiles, then Theorem 10.2 combined with the exact uniform per-step KL approximation (Section 5.3) suggests a much smaller cover width (about $w \approx 17$ in the same $T = 50, m = 20$ setting). This gap is expected: MI/Fano controls *average* information, while Theorem 10.1 controls worst-case posteriors transcript-by-transcript. A short script that computes the smallest feasible w for given $(T, m_t, n, \delta_{T+1}, \eta)$ is included as `compiler_design_worksheet_latest.py`.

Figure 3 (worksheet scaling). The posterior-first design worksheet in Section 10.4 implies that the required equalization fanout w scales roughly like $\sum_t m_t$ divided by the allowable posterior slack $\ln(1 + \eta) - \varepsilon_{\text{stop}}$, up to the exact nonlinearity $\log \frac{1+a}{1-a}$. The plot below shows w vs η for the two schedules used in the worked examples (uniform $m = 20, T = 50$; and mixed $m \in \{8, 32\}, T = 100$), comparing the loose per-step ratio $a_t = m_t/w$ to the canonical injection ratio $a_t = (m_t - 1)/w$.

Text-only replacement for Fig. 3. The plot is omitted (text-only submission). It visualized the design worksheet curve: the minimal cover width w needed to hit posterior target $(1 + \eta)/K$ decreases with η , and is typically much larger under transcript-by-transcript posterior control than under MI/Fano. (To regenerate the original plot, run `make_figures_v67.py`.)

13 11. Related work (non-exhaustive)

This paper is intentionally modular: we treat metadata leakage channels (stop time and trace/marks) as *interfaces* that export quantitative contracts, so that compilers can be analyzed by composition rather than by a monolithic, system-level argument. Closest in spirit are works that (i) give operationally meaningful leakage measures, and (ii) provide contraction/perturbation tools for sequential observation processes.

Our analysis draws on (a) pointwise leakage measures that directly control posteriors, (b) strong data-processing and contraction coefficients for Markovian observation processes, and (c) sharp KL-TV inequalities used to turn TV-style compiler contracts into KL/MI bounds. The following anchors touch the specific tools used here:

- **Pointwise leakage / maximal leakage.** Operational formulations and closed forms for maximal leakage and related pointwise measures appear in [IWM18] (which motivates our Section 7 “posterior-first” stance).
- **SDPIs and contraction coefficients.** Strong data-processing inequalities and contraction viewpoints aligned with our “reference process + contraction” accounting are developed in [PW15].
- **Doebelin/Dobrushin-type coefficients.** Structural relationships among Doebelin-style measures and contraction coefficients appear in [MS23], supporting the Section 9 mixing/contraction bookkeeping.

- **Perturbation sensitivity of Markov chains.** Quantitative stability bounds for uniformly ergodic chains under kernel perturbations appear in [Mit05], conceptually close to our “equalization as controlled perturbation” lens.
- **Reverse Pinsker and KL–TV envelopes.** Reverse Pinsker-type inequalities and related bounds appear in [Sas15], while Bretagnolle–Huber [BH79] and a modern KL–TV note [Can22] motivate our Lemma 10.3 / Section 8 envelope discussion.
- **Background information theory.** Standard identities and inequalities (chain rules, Fano, etc.) are in [CT06].

14 12. Open problems (updated)

1. **Global optimal stop-time padding beyond mixtures.** Kernel-TV-Pad gives the true optimum among kernel-feasible paddings, and we exhibit separating families. What is the best *structural* characterization of optimal kernels (e.g., sparsity patterns, monotonicity, bounds on the number of randomized rows)? Concrete progress would include: (i) a theorem that mixtures achieve a constant-factor approximation to the LP for all instances; or (ii) an efficiently checkable condition under which the LP optimum *is* a mixture; or (iii) lower-bound constructions that force multi-row randomization in a quantified way.
2. **Deriving history-uniform contracts from real retry/repair policies.** The compiler calculus consumes per-history TV (or pointwise) deviation bounds. Can we systematically derive such bounds from concrete overlay protocols (timeouts, retries, backoff, batching, repair), especially when the cover kernel is state-dependent? Progress would mean reusable lemmas of the form: “Policy X implies $\text{TV}(P_t(\cdot \mid c, h), K_t(\cdot \mid h)) \leq \delta_t$ for all histories h ” with explicit δ_t as a function of protocol parameters.
3. **Moment-based sequential KL/MI accounting without global floors.** Lemma 5.2 avoids a global $q_{\min}$ for one-step KL via moments. Can analogous moment-style bounds propagate through sequential composition (Theorem 6.1 / 10.2) to avoid global $k_{\min}$ assumptions, especially when observation alphabets are large and supports drift with history? Progress would mean MI/KL bounds stated in terms of per-step moments (or collision probabilities) that remain non-vacuous at scale.
4. **Tightness of near-1/ K posterior bounds under scheduling + injection.** The sequential PML/posterior theorems yield clean sufficient conditions, but can be loose. When are the bounds tight (up to constants), and when do they blow up unnecessarily? Progress would include matching lower bounds (adversary strategies) for scheduled-contact mechanisms, or identifying symmetric regimes where an exact posterior analysis is possible.

15 References

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- [PW15] Yury Polyanskiy and Yihong Wu. *Strong data-processing inequalities for channels and Bayesian networks*. arXiv:1508.06025 (2015).

- [MS23] Anuran Makur and Japneet Singh. *Doebelin Coefficients and Related Measures*. arXiv:2309.08475 (2023). Journal-ref: IEEE Transactions on Information Theory, 70(7), July 2024.
- [Sas15] Igal Sason. *On Reverse Pinsker Inequalities*. arXiv:1503.07118 (2015).
- [Can22] Clément L. Canonne. *A short note on an inequality between KL and TV*. arXiv:2202.07198 (2022).
- [BH79] Jean Bretagnolle and Catherine Huber. *Estimation des densités: risque minimax*. Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete, 47:119–137, January 1979. DOI: 10.1007/BF00535278.
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16 Appendix A. Pipeline diagram and module pseudocode

This appendix provides the promised one-page *compiler calculus* pipeline diagram (Figure 1) and lightweight pseudocode for the core modules. The intent is operational clarity: each box exports an explicit contract that the composition theorems (Theorems 10.1–10.2) consume. **Figure 1 (Compiler calculus pipeline).**

```
User/profile C Public time/schedule
| |
| (payload layer hides | PSC: public schedule J_t
| contents; adversary | (action set per slot)
| sees metadata) v
| Allowed action set J_t
| |
| | MC-EQ: choose w actions within J_t
| | (1 real + w-1 covers)
v v
Real target action Observed mark O_t in J_t
| |
| | Contract exported:
| | (TV)  $TV(P_t(\cdot|c,h), K_t(\cdot|h)) \leq 1/w$ 
| | (L $\infty$ )  $|P_t(y|c,h) - K_t(y|h)| \leq 1/w$ 
| v
| Mark transcript O_{1:T}
|
| Base success time S_c
| (secret-dependent)
v
StopTimePad: kernel-feasible padding  $\Phi(t|s)$ 
|
| Output: padded stop time  $\tilde{T}$ 
| Contract exported (two options):
| (TV)  $TV(P_c, P_{\{c'\}}) \leq \tau$  (Kernel-TV-Pad)
| (post) additive deviation + floor vs reference R
v
Joint transcript (O_{1:T},  $\tilde{T}$ )
|
```

v

Composition theorem:

- Theorem 10.1: posterior control (near $1/K$)
- Theorem 10.2: MI/Fano control

16.1 A.1 PSC: public scheduling

PSC publishes the action-set schedule J_t and enforces fixed-size traffic to every shard in J_t each slot. This collapses shard-level marks: the adversary learns J_t from public time, and only learns the *within-set* action identifier from O_t .

16.2 A.2 MC-EQ: mixture equalization within a scheduled set

Below is a reference implementation sketch for one slot.

Input: history h , public set J_t , cover kernel $K_t(\cdot|h)$ on J_t , $w \geq 1$

1. Sample cover multiset COV of size $w-1$ i.i.d. from $K_t(\cdot|h)$
2. Sample real action A_{real} from the client rule (may depend on secret and h)
3. Send one request to each action in $\{A_{\text{real}}\} \cup \text{COV}$ (duplicates allowed)
4. Output mark O_t as the multiset or the (canonicalized) contacted identifier

Contract. For the induced distribution of the observable mark O_t (after canonicalization), $P_t(\cdot | C = c, H_{t-1} = h)$ is a $\rho = 1/w$ injection mixture relative to $K_t(\cdot | h)$, hence: - $\text{TV}(P_t(\cdot | c, h), K_t(\cdot | h)) \leq 1/w$, and - $|P_t(y | c, h) - K_t(y | h)| \leq 1/w$ for all $y \in J_t$ (Lemma 4.2). These are precisely the per-slot hypotheses consumed by Theorem 10.1.

16.3 A.3 StopTimePad: kernel-feasible padding

A kernel-feasible stop-time padding is a lower-triangular stochastic matrix $\Phi(t | s)$ supported on $t \geq s$.

Input: realized success time s , padding kernel Φ

1. Sample padded stop time $\tilde{T} \sim \Phi(\cdot|s)$
2. Run dummy traffic until time $\tilde{T}$ and then terminate

Output: $\tilde{T}$

Two useful exported contracts are: - **Pairwise-TV contract** (Kernel-TV-Pad): for all secrets c, c' , $\text{TV}(P_c, P_{c'}) \leq \tau$. - **Posterior-friendly contract** (PosteriorStopTime-Pad): existence of a reference R with floor κ_T and per-time additive deviation δ_T , so that $|P_c(t) - R(t)| \leq \delta_T$ and $R(t) \geq \kappa_T$ for all t . This is a direct plug-in to Theorem 10.1 (via Proposition 2.15).

16.4 A.4 How the contracts compose

Theorem 10.1 treats the stop time $\tilde{T}$ as one more observation step with reference R . Theorem 10.2 instead converts per-step TV deviations to per-step KL (Lemma 10.3) and sums KL via the chain rule before applying Fano.

17 Appendix B. Numerical sanity checks (non-adversarial)

These checks are **not** part of the formal proof. They are quick Monte Carlo tests intended to catch algebra/constant mistakes and to gauge looseness.

1) **Moment KL bound for one-real-among- w injection (Lemma 5.2).**

We sample a dimension m uniformly from $\{2, \dots, 40\}$, a base distribution q uniformly from the simplex (Dirichlet(1)), an index j uniformly from $\{1, \dots, m\}$, and w uniformly from $\{2, \dots, 50\}$. With $\rho = 1/w$ and $p = (1 - \rho)q + \rho e_j$, we check

$$D_{\text{KL}}(p\|q) \leq \rho^2 \left(\frac{1}{q_j} - 1 \right).$$

Over 5000 trials we observed **no violations**. The worst observed ratio $D_{\text{KL}}/\text{bound}$ was about 0.595.

2) **KL→TV envelope numbers (Pinsker vs Bretagnolle–Huber).**

The script prints a small table for $\varepsilon \in \{0.01, 0.03, 0.05, 0.1, 0.2, 0.3\}$ comparing Pinsker’s $2\varepsilon^2$ to Bretagnolle–Huber’s $-\ln(1 - \varepsilon^2)$ and their ratio.

The script used is `sanity_sims_v57.py` (invoked by the one-command runner in Appendix C.3).

18 Appendix C. Reproducibility checklist

This draft ships with small scripts that reproduce the main numerical/certificate claims. All scripts are deterministic given their hard-coded instances and should run in seconds on a laptop.

18.1 C.1 Stop-time padding

- **Binary separation (kernel-feasible beats mixtures):** `stop_time_dual_cert_v16.py` (primal kernel + dual multipliers + KKT residuals).
- **Multi-row optimal padding example:** `stop_time_multirow_cert_v47.py` (two-row randomization instance; HiGHS KKT residuals).
- **Parametric family (closed form, linear-in-horizon gap):** `stop_time_dual_cert_v18.py` (checks dual inequalities for the family and confirms the closed-form objective matches the LP).
- **K -ary star vs pairwise gap (Proposition 2.12):** `stop_time_star_gap_v21.py` (prints optimal kernels/objectives), and `stop_time_kary_cert_v38.py` (extracts HiGHS dual multipliers and checks KKT).
- **PosteriorStopTime-Pad certificate (Problem 2.14):** `stop_time_posterior_cert_v46.py` (HiGHS primal–dual gap and KKT residuals for a small instance).
- **TV geometry (Lemma 2.13):** `tv_radius_geometry_v24.py`.

18.2 C.2 Trace / posterior / design worksheet

- **Posterior-composition sanity simulation** (for Theorem 7.2 / 10.1 style bounds): `compiler_calculus_sims_v`
- **Exact-vs-bounding KL accounting for injection:** `w_compare_bounds.py`.
- **Design worksheet (solve for w under posterior targets and MI/Fano contrast):** `compiler_design_worksheet_latest.py`.

18.3 C.3 One-command runner

- A convenience runner is provided as `reproduce_all_v67.sh`.

18.4 C.4 Figures

- **Figure 2 (gap vs horizon)** and **Figure 3 (worksheet scaling)**: `make_figures_v67.py` (writes `fig_gap_vs_n` and `fig_w_vs_eta`).

19 Appendix D. BibTeX entries (copy/paste)

The following BibTeX entries cover all citation tags used in the draft. They are intended as a convenience for submission; feel free to replace with your preferred bibliography style.

```
@book{CT06,
  author    = {Thomas M. Cover and Joy A. Thomas},
  title     = {Elements of Information Theory},
  edition   = {2},
  publisher = {Wiley-Interscience},
  year      = {2006},
  isbn      = {978-0471241959}
}

@misc{IWM18,
  author      = {Ibrahim Issa and Aaron B. Wagner and Sudeep Kamath},
  title       = {An Operational Approach to Information Leakage},
  year        = {2018},
  eprint      = {1807.07878},
  archivePrefix= {arXiv},
  primaryClass = {cs.IT}
}

@misc{PW15,
  author      = {Yury Polyanskiy and Yihong Wu},
  title       = {Strong data-processing inequalities for channels and Bayesian networks},
  year        = {2015},
  eprint      = {1508.06025},
  archivePrefix= {arXiv},
  primaryClass = {cs.IT}
}

@misc{MS23,
  author      = {Anuran Makur and Japneet Singh},
  title       = {Doebelin Coefficients and Related Measures},
  year        = {2023},
  eprint      = {2309.08475},
  archivePrefix= {arXiv},
  primaryClass = {cs.IT}
}
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}

```
@article{Mit05,  
  author = {A. Yu. Mitrophanov},  
  title  = {Sensitivity and convergence of uniformly ergodic {M}arkov chains},  
  journal = {Journal of Applied Probability},  
  volume = {42},  
  number = {4},  
  pages  = {1003--1014},  
  year   = {2005},  
  doi    = {10.1239/jap/1134587812}  
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@misc{Can22,  
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  title       = {A short note on an inequality between {KL} and {TV}},  
  year        = {2022},  
  eprint      = {2202.07198},  
  archivePrefix = {arXiv},  
  primaryClass = {cs.IT}  
}
```

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@misc{Sas15,  
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  title       = {On Reverse Pinsker Inequalities},  
  year        = {2015},  
  eprint      = {1503.07118},  
  archivePrefix = {arXiv},  
  primaryClass = {cs.IT}  
}
```

```
@incollection{BH79,  
  author      = {Jean Bretagnolle and Catherine Huber},  
  title       = {Estimation des densités: risque minimax},  
  booktitle   = {Séminaire de Probabilités {XII} (Univ. Strasbourg, Strasbourg, 1976/1977)},  
  series      = {Lecture Notes in Mathematics},  
  volume      = {649},  
  pages       = {342--363},  
  publisher   = {Springer},  
  address     = {Berlin},  
  year        = {1978}  
}
```